

What We Did This Session

A simple, no-jargon walkthrough of this session's conversation - what was asked, and what happened

6 September 2026

Two parts today. First, properly saving all our work under real version control for the first time - and catching a real security slip along the way. Second, three more real experiments chasing the very last open question on the cross-border trading work: a storage test, a single-country test, and a full 10-country build that hit and fixed a genuine software bug in AMIRIS itself.

1. Setting Up Real Version Control - and Catching a Real Slip

You asked: "Commit these changes."

Found the project had never actually been under its own version control - and stopped before committing a real credential by mistake.

Checking first (rather than just running the commit) turned up two real problems: the version-control history on this computer covered the ENTIRE home folder, not just this project, mixing in unrelated personal files - and partway through preparing the commit, spotted that the file holding our real API access tokens (ENTSO-E, renewables.ninja) was about to be saved into that history by mistake. Stopped, excluded it properly, and set up a clean, separate, dedicated history just for this project instead - the safer choice given what the first option risked.

Fixed the author details too, once flagged.

The save was initially credited to the wrong name/email (a leftover default on this shared computer). Corrected it to the right one and confirmed the save now shows the right authorship - since this was the very first save in a brand new history, fixing it in place was clean and safe, with nothing else to disturb.

Clarified after a follow-up question: this save exists only on this computer for now, not on GitHub - nothing has been uploaded anywhere yet. That's a separate, deliberate next step whenever you're ready for it.

2. One More Lead on the Last Open Question

You asked: "Continue building out the export module improvements."

Tested directly whether the other side's own storage was the real limiter - and ruled it out.

The one open thread left: a small number of hours where Germany imports less than it could, even with room to spare. Checked directly whether the rest of Europe's own batteries and reservoirs were simply empty during those hours. They weren't - neither ever came close to its own limit, in either year checked. Rules out 'nothing left to give' as the explanation, and points more firmly at the trading engine's own internal logic as the real cause - not yet fully provable without a deeper software change.

3. Pushing Further: One Real Country Instead of the Combined Zone

You asked: "What of building for each of the countries, maybe we should try that?"

Flagged the real risk first, then built a single-country test as a fair first check.

Warned directly that splitting every country out separately, the same simple way, could make things WORSE rather than better, based on what the earlier work had already shown. You agreed a single-country test first was the sensible way to check that before committing to the bigger build. Built France properly as its own real zone - the largest economy involved, with enough real data already on hand to do it right.

Result: a real step backward, with a clear, understood reason.

The 'ran out of power' hours jumped sharply (7 to 167) once France was split out. Checked why rather than leaving it a mystery: France's own huge nuclear fleet had been quietly propping up the WHOLE combined zone's supply. Pulled out on its own, the remaining combined zone lost that cushion, while France's own leftover cheap power could no longer reach it either, stuck behind a single narrow border link to Germany.

4. Going All the Way: All 10 Real Countries at Once

You asked: "Keep digging into the Denmark bug first."

Built all 10 real neighbouring countries as their own zones - and hit a genuine software bug.

Since one country made things worse for a clear reason, the natural next test was whether the SAME problem compounds across all 10, or whether something different happens at full scale. The build itself crashed - and this time, unlike every earlier problem in this project, the cause wasn't obvious from checking the settings.

Tracked it down properly rather than guessing: isolated it to Denmark specifically.

Tested one country at a time. Austria worked perfectly. Denmark, tested the same way, crashed every time. Checked Denmark's real numbers: its actual hydropower is only 7 megawatts - thousands of times smaller than every other country involved. Proved this was the real cause, not a coincidence, by testing it twice in each direction: the exact same setup with just the numbers made bigger ran perfectly, every time; the exact same setup with the real tiny numbers crashed, every time. Also tried simply leaving Denmark's hydropower out entirely - that did NOT fix it either, so it had to be properly patched, not just removed.

Fixed it honestly: a clearly labelled technical workaround, not invented data.

Applied a documented, openly-artificial minimum size to Denmark's (and the similarly-tiny Netherlands') hydropower, purely so the software could run - clearly marked in the file itself as NOT representing either country's real hydropower scale. With that fix in place, the full 10-country build ran cleanly all the way through.

	Combined zone (best)	France alone	All 10 countries
'Ran out of power' hours	7	167	120
Average price gap, ordinary days	-2.90	+9.65	+4.87
Match quality (trustworthiness)	0.717	0.716	0.731 (best yet)

Result: better than expected, but still not the winning approach.

Genuinely surprising: going all the way to 10 real countries did BETTER than splitting out just one - likely because Germany can now draw on many countries' cheap power at once, instead of being stuck behind a single narrow link. But it still falls short of the original combined-zone approach on the two measures that matter most. Decision: keep the combined zone as our best, standing result - both new builds are kept as real, documented evidence, not deleted.

5. Bringing Both Documents Up to Date

You asked: "Update the supervisor deck with Phase 43. / Update the daily session-recap PDF too."

Both the supervisor deck and the technical progress report now reflect all of today's real work.

Added five new slides to the supervisor deck covering the storage test, the France result, the Denmark bug story, and the final 10-country comparison table - placed right after the earlier market-coupling section and before the closing summary slides. The technical progress report already had the full written record of all three experiments from earlier in the session.